DTC U0028:87 [BLIND SPOT MONITORING (BSM) SYSTEM]

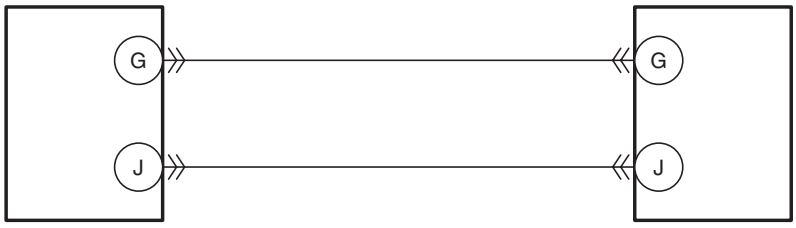
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Caution

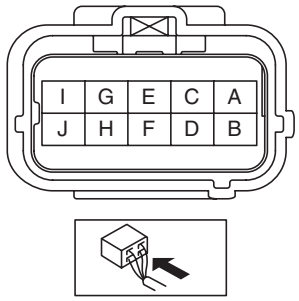
- **When replacing a blind spot monitoring (BSM) control module (RH) having any of the following part numbers, it is necessary to replace the blind spot monitoring (BSM) control module (LH) at the same time to perform the configuration normally.**
  - **— DA6A-67Y80 A**
  - **— DA6A-67Y80 B**

Description	Blind spot monitoring (BSM) control module (LH) and (RH) communication error
Detection condition	• No communication between blind spot monitoring (BSM) control modules (LH) and (RH) for **1 s or more** with the ignition switched ON (engine off or on).
Fail-safe	• Blind spot monitoring (BSM) system is stopped.
Possible cause	• Blind spot monitoring (BSM) control module (RH) connector or terminal malfunction • Blind spot monitoring (BSM) control module (LH) connector or terminal malfunction • Short to ground in wiring harness between the following terminals: — Blind spot monitoring (BSM) control module (LH) terminal G—Blind spot monitoring (BSM) control module (RH) terminal G — Blind spot monitoring (BSM) control module (LH) terminal J—Blind spot monitoring (BSM) control module (RH) terminal J • Short to power supply in wiring harness between the following terminals: — Blind spot monitoring (BSM) control module (LH) terminal G—Blind spot monitoring (BSM) control module (RH) terminal G — Blind spot monitoring (BSM) control module (LH) terminal J—Blind spot monitoring (BSM) control module (RH) terminal J • Open circuit in wiring harness between the following terminals: — Blind spot monitoring (BSM) control module (LH) terminal G—Blind spot monitoring (BSM) control module (RH) terminal G — Blind spot monitoring (BSM) control module (LH) terminal J—Blind spot monitoring (BSM) control module (RH) terminal J • Blind spot monitoring (BSM) control module (RH) malfunction • Blind spot monitoring (BSM) control module (LH) malfunction

BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) BLIND SPOT MONITORING (BSM) CONTROL MODULE (RH)



BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH)/(RH)
WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT BLIND SPOT MONITORING (BSM) CONTROL MODULE (RH) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the blind spot monitoring (BSM) control module (RH) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
2	**INSPECT BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) CONNECTOR CONDITION** • Disconnect the blind spot monitoring (BSM) control modules (LH) connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.
3	**INSPECT CAN COMMUNICATION CIRCUIT BETWEEN BLIND SPOT MONITORING (BSM) CONTROL MODULES (LH) AND (RH) FOR SHORT TO GROUND** • Verify that the blind spot monitoring (BSM) control module (LH) and (RH) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Blind spot monitoring (BSM) control module (RH) terminal G — Blind spot monitoring (BSM) control module (RH) terminal J • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Blind spot monitoring (BSM) control module (LH) terminal G—Blind spot monitoring (BSM) control module (RH) terminal G • Blind spot monitoring (BSM) control module (LH) terminal J—Blind spot monitoring (BSM) control module (RH) terminal J **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 6.
		No	Go to the next step.
4	**INSPECT CAN COMMUNICATION CIRCUIT BETWEEN BLIND SPOT MONITORING (BSM) CONTROL MODULES (LH) AND (RH) FOR SHORT TO POWER SUPPLY** • Verify that the blind spot monitoring (BSM) control module (LH) and (RH) connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the following terminals (wiring harness-side): — Blind spot monitoring (BSM) control module (RH) terminal G — Blind spot monitoring (BSM) control module (RH) terminal J • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Blind spot monitoring (BSM) control module (LH) terminal G—Blind spot monitoring (BSM) control module (RH) terminal G • Blind spot monitoring (BSM) control module (LH) terminal J—Blind spot monitoring (BSM) control module (RH) terminal J **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 6.

Step	Inspection		Action
5	**INSPECT CAN COMMUNICATION CIRCUIT BETWEEN BLIND SPOT MONITORING (BSM) CONTROL MODULES (LH) AND (RH) FOR OPEN CIRCUIT** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the blind spot monitoring (BSM) control module (LH) and (RH) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Blind spot monitoring (BSM) control module (LH) terminal G—Blind spot monitoring (BSM) control module (RH) terminal G — Blind spot monitoring (BSM) control module (LH) terminal J—Blind spot monitoring (BSM) control module (RH) terminal J • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Blind spot monitoring (BSM) control module (LH) terminal G—Blind spot monitoring (BSM) control module (RH) terminal G • Blind spot monitoring (BSM) control module (LH) terminal J—Blind spot monitoring (BSM) control module (RH) terminal J **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
6	**VERIFY IF MALFUNCTIONING LOCATION IS BLIND SPOT MONITORING (BSM) CONTROL MODULES (RH) DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the blind spot monitoring (BSM) control module using the M-MDS. (See CLEARING DTC [BLIND SPOT MONITORING (BSM) SYSTEM].) • Switch the ignition ON (engine off or on) and wait for **2.5 s or more**. • Retrieve the blind spot monitoring (BSM) control module DTCs using the M-MDS. (See DTC INSPECTION [BLIND SPOT MONITORING (BSM) SYSTEM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the blind spot monitoring (BSM) control module. (See BLIND SPOT MONITORING (BSM) CONTROL MODULE REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to Step 8.
7	**VERIFY IF MALFUNCTIONING LOCATION IS BLIND SPOT MONITORING (BSM) CONTROL MODULES (LH) DEPENDING ON REPEATABILITY** • Clear the DTC for the blind spot monitoring (BSM) control module using the M-MDS. (See CLEARING DTC [BLIND SPOT MONITORING (BSM) SYSTEM].) • Switch the ignition ON (engine off or on) and wait for **2.5 s or more**. • Retrieve the blind spot monitoring (BSM) control module DTCs using the M-MDS. (See DTC INSPECTION [BLIND SPOT MONITORING (BSM) SYSTEM].) • Is the same DTC displayed?	Yes	Replace the blind spot monitoring (BSM) control module (LH), then go to the next step. (See BLIND SPOT MONITORING (BSM) CONTROL MODULE REMOVAL/INSTALLATION.)
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BLIND SPOT MONITORING (BSM) SYSTEM].)
		No	DTC troubleshooting completed.